Putting Numbers on the Phenomena

SYNOPSIS

In **N2** you watched things happen and said which law was responsible. That is half of physics. The other half is *how much* — and it is the half that decides whether a bridge stands up. This note goes back to the same demonstrations and works out the numbers: what the penny weighs, how hard each person on a trolley accelerates, and where to point the launcher. The physics is not new. What is new is the discipline of getting an answer you can defend.

BEFORE YOU READ ON — PREDICT

You are on a trolley with a total mass of **100 kg**. You push on someone else whose trolley and body come to **200 kg**. Two separate questions, and they do not have the same answer:

Do the two of you feel the **same size force**?

Do the two of you get the **same acceleration**?

Answer both, yes or no, with one sentence each, before you read on.

PERFORMANCE EXPECTATIONS

1. **Convert** a mass given in grams to kilograms before substituting it, and **show** the conversion as a step rather than doing it in your head.
2. **Carry** units through every line of a calculation and **state** the unit of the answer. An answer without a unit is not an answer.
3. **Compute** the weight of a mass from w = mg.
4. **Apply** a = F/m to each member of an interacting pair, and **explain** why the two accelerations differ although the two forces do not.
5. **Use** trigonometry to find a launch angle from a height and a distance.
6. **Check** whether an answer is reasonable before writing it down, and say what made you accept it.

MOVEMENT 1 — THE RULES OF THE ROAD

How to get an answer you can defend

EVERY TIME

convert → substitute → carry the units → check it is reasonable

MOVEMENT 2 — WHAT THE PENNY WEIGHS

From mass to weight

Eq. 1 $w = F_g = mg$

g = 9.8 m/s² near the Earth's surface

TASK 1 — A PENNY

1. The mass of a penny is **2.5 g**. Convert it to kilograms, and write the conversion as its own line.

2. Find its weight. Show the substitution with units attached.

3. Is the answer reasonable? Say what you compared it with.

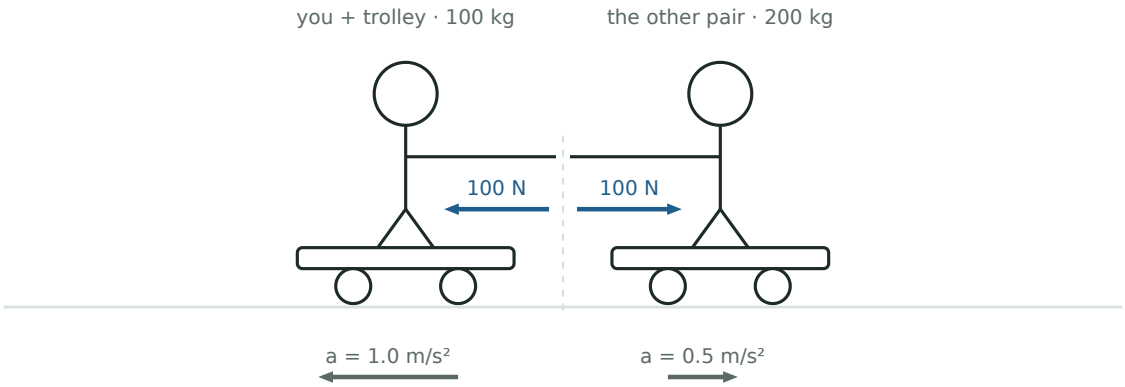
TASK 2 — AN ADULT

1. A typical adult has a mass of **70 kg**. Find the weight, with units on every number.

2. Your answer is a number of newtons. Roughly how many pounds is that, and does the comparison make sense to you?

MOVEMENT 3 — THE THIRD LAW, WITH NUMBERS

Equal forces, unequal accelerations



the force arrows match. the acceleration arrows do not.

TASK 3 — BOTH SIDES OF THE PUSH

1. You and your trolley come to 100 kg. You push with 100 N. Find your acceleration.

2. The other person and their trolley come to 200 kg. What force acts on *them*? State it before you calculate anything.

3. Find their acceleration.

4. Compare the two accelerations. Write the comparison as a ratio, and say what that ratio is the ratio *of*.

5. Go back to the wall in N2 and finish the argument properly: name the quantity that makes the wall's response invisible, and say what would have to be true for you to see the wall move.

6. In **N1** you met the claim that velocity is relative but acceleration is absolute. You and the other person disagree about who is moving. Do you disagree about the *accelerations*? Use your two numbers to answer.

WITH CALCULUS

$\Sigma F = ma$ is not the most general form of the second law. Newton wrote it about **momentum**, $p = mv$:

$$\Sigma F = \mathrm{d}p / \mathrm{d}t$$

Which turns into the equation you have been using all note, but only after an assumption that nothing on this page has stated out loud.

1. Substitute $p = mv$ and differentiate the **product**. You should get two terms.

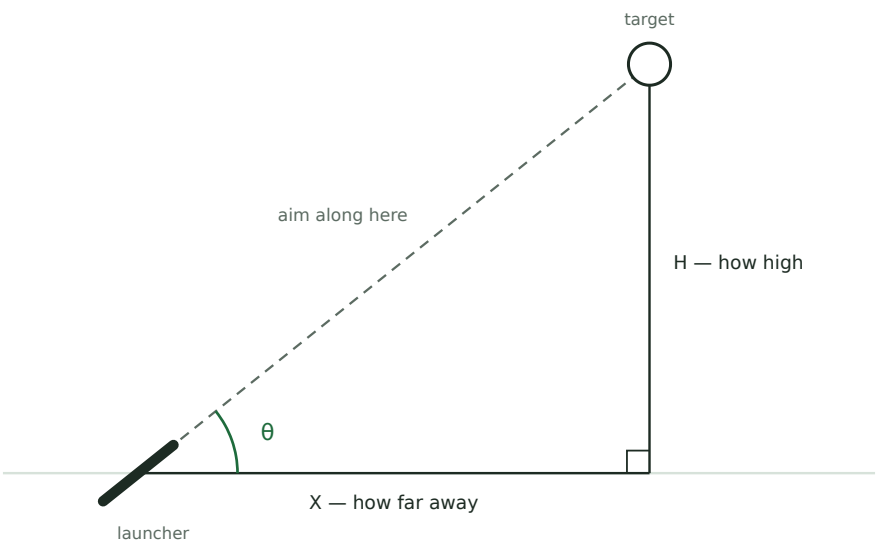
2. One of them is ma. What is the other, and under what condition is it zero?

3. That condition is the assumption. It held in every calculation in this note — say why, and then name a situation in which it would **fail**.

4. Back to **N1**. You showed there that two inertial observers agree about a. Do they agree about p? Do they agree about $\mathrm{d}p/\mathrm{d}t$? Which of those three answers is the reason $\Sigma F = \mathrm{d}p/\mathrm{d}t$ can be a law at all?

MOVEMENT 4 — AIMING THE LAUNCHER

Where to point the barrel



TASK 4 — THE LAUNCH ANGLE

1. Draw the setup in your own notes and label X, H and θ.

2. Which pair of sides does θ sit between, and therefore which trigonometric ratio do you need? Write the ratio before putting any numbers in.

3. With X = 2.5 m and H = 2.0 m, find θ.

4. Check it: is your angle less than or greater than 45°, and does that match the picture?

BACK TO YOUR PREDICTION

You answered two questions at the top: whether the forces are equal, and whether the accelerations are equal. Which of your two answers was right? If you gave the same answer to both, name the step in Task 3 that separates them.

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